

An Optimised *Ex Vivo* Porcine Retinal Explant Model to Evaluate the Therapeutic Potential of Rosmarinic Acid-Releasing Contact Lenses

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Abstract

Therapeutic management of diabetic retinal diseases remains challenging. Current treatments often require repeated intravitreal injections, which are invasive and burdensome, while topical eye drops provide limited drug delivery to the retina due to rapid clearance and poor penetration. Drug-loaded contact lenses (CLs) have emerged as a promising alternative. By maintaining prolonged contact with the ocular surface, they can enable sustained and controlled drug release, potentially improving bioavailability and reducing the need for frequent invasive interventions. Rosmarinic acid (RA) is a bioactive polyphenol with recognized antioxidant, anti-inflammatory, and neuroprotective effect. These pharmacological properties make it a promising therapeutic candidate for the treatment of diabetic retinopathy. In a previous work a silicon-based hydrogel was formulated to be used in CLs for the vehiculation of RA¹. It was able to ensure the release of the drug for 24 h under hydrodynamic conditions which mimic those found in the eye. A mathematical model was applied to predict the concentrations achieved in the retina upon the application of these CLs.

Here, we applied an optimized organotypic *ex vivo* model using porcine retinal explants to examine the neuroprotective impact of RA at those concentrations. Retinal explants were cultured for 48 hours in medium containing RA (86-400 ng/mL). After incubations, samples were cryosectioned and subjected to immunofluorescence staining. Survival of retinal ganglion cells (RGCs) was evaluated using Brn3a as a selective marker.

Exposure to RA led to a marked preservation of RGCs density compared with controls, which demonstrated degenerative alterations. Quantitative measurements were obtained through fluorescence microscopy and subsequent image analysis was done using Fiji/ImageJ software.

Overall, these findings support the therapeutic promise of RA-releasing CLs for diabetic eye complications and confirm the suitability of porcine retinal explants as a robust model for testing novel ocular drug-delivery approaches.

1. Centeno Duarte A, Toffoletto N, Martins Pais R, et al. Rosmarinic acid-eluting contact lenses as a multifunctional therapeutic system for diabetic ocular complications. *Int J Pharm.* 2026;691:126591. doi:10.1016/j.ijpharm.2026.126591

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